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ORIGINAL ARTICLE

Effect of different work and recovery settings during high-intensity intermittent training on maximal oxygen uptake and session volume responses

Effet de différents paramètres de travail et de la récupération pendant l'entraînement intermittent à haute intensité sur l'absorption maximale d'oxygène et le volume de la séance

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KEYWORDS

Interval training;
Oxygen consumption;
Amateur runners;
Time to exhaustion

Summary

Objectives. – The aim of the present study was to verify the training session volume and the time spent at or above 90% of maximal oxygen uptake ($t_{90VO_{2max}}$) under different work and recovery settings during high-intensity intermittent training.

Equipment and methods. – After having the maximal oxygen uptake (VO_{2max}) and peak velocity (V_{peak}) determined, eleven male runners (age: 25.45 ± 4.13 years; VO_{2max} : 55.88 ± 1.43 ml.kg⁻¹.min⁻¹ and personal best in urban 5 km: 18 min: 30s $\pm$ 01 min: 25s) completed three short HIIT (30s/30s, 40s/40s e 40s/20s) at 110% of V_{peak} until exhaustion. The time sustained in oxygen fractions above 90% of VO_{2max} , total exercise time (TT), number of complete series performed (NS) and time of exercise (TE) were quantified and compared between the different sessions.

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MOTS CLÉS

Entraînement par intervalles ;
Consommation d'oxygène ;
Coureurs amateurs ;
Temps à épuisement

Results. – The $t90VO_{2max}$ was not different between sessions ($P > 0.05$); in turn, the 30s/30s protocol was greater than both 40s/40s and 40s/20s at the volume variables (30s/30s, TT (s): 1463 ± 86.14 , NS (n): 24.18 ± 1.48 and TE (s): 737.36 ± 41.9 –40s/40s, TT (s): 1043 ± 69.39 , NS (n): 12.91 ± 0.87 and TE (s): 526.45 ± 34.72 –40s/20s, TT (s): 415.55 ± 30.08 , NS (n): 6.72 ± 0.51 and TE (s): 281 ± 20.01). For the prescription of training sessions, the 30s/30s protocol being the most appropriate when there is the goal to increase the volume session. On the other hand, protocol 40s/20s may optimize training time when the purpose is to stimulate high oxygen fractions, as its results are like those of other protocols in $t90VO_{2max}$ with time reduced up to four times.

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Résumé

Objectifs. – Le but de la présente étude était de vérifier le volume de la séance d'entraînement et le temps passé à 90 % ou plus d'absorption maximale d'oxygène ($t90VO_{2max}$) dans différents contextes de travail et de récupération pendant l'entraînement intermittent à haute intensité (HIIT).

Équipement et méthodes. – Après avoir déterminé l'absorption maximale d'oxygène (VO_{2max}) et la vitesse maximale (V_{peak}), onze coureurs masculins (âge : $25,45 \pm 4,13$ ans ; VO_{2max} : $55,88 \pm 1,43$ mL.kg⁻¹.min⁻¹ et meilleur résultat personnel urbain 5 km : 18 min : 30s ± 01 min : 25s) complété trois courts HIIT (30s/30s, 40s/40s et 40s/20s) à 110 % de V_{peak} jusqu'à épuisement. Le temps soutenu dans les fractions d'oxygène au-dessus de 90 % de VO_{2max} , le temps total d'exercice (TT), le nombre de séries complètes réalisées (NS) et le temps d'exercice (TE) ont été quantifiés et comparés entre les différentes sessions.

Résultats. – Le $t90VO_{2max}$ n'était pas différent entre les séances ($p > 0,05$) ; à leur tour, le protocole 30s/30s était plus grand que les 40s/40s et 40s/20s aux variables de volume (30s/30s, TT (s) : $1463 \pm 86,14$, NS (n) : $24,18 \pm 1,48$ et TE (s) : $737,36 \pm 41,9$ –40s/40s, TT (s) : $1043 \pm 69,39$, NS (n) : $12,91 \pm 0,87$ et TE (s) : $526,45 \pm 34,72$ –40s/20s, TT (s) : $415,55 \pm 30,08$, NS (n) : $6,72 \pm 0,51$ et TE (s) : $281 \pm 20,01$). Pour la prescription des sessions de formation, le protocole 30s/30s est le plus approprié lorsque l'objectif est d'augmenter la session de volume. D'autre part, le protocole 40s/20s peut optimiser le temps d'entraînement lorsque l'objectif est de stimuler des fractions d'oxygène élevées, car ses résultats sont similaires à ceux d'autres protocoles dans $t90VO_{2max}$ avec un temps réduit jusqu'à quatre fois.

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1. Introduction

The maximal oxygen uptake (VO_{2max}) is, along with running economy and the physiological transition thresholds (lactate and ventilatory), a great contributor to the performance success in middle and long-distance running [1,2]. Thus, sports science has sought to develop training methods aiming to increase this measure, as subtle raisings of this variable could result in substantial gains and help athletes succeed in competitions [2,3].

In fact, there is scientific consensus that increasing the time spent at or above 90% of VO_{2max} during training sessions is fundamental for this variable to be modified. Such condition ($\geq 90\% VO_{2max}$) may be reached and kept for longer periods during high-intensity exercises. In this manner, the high-intensity intermittent training (HIIT) method has become one of the best alternatives to coaches, researchers and athletes intending to stimulate VO_{2max} alteration [3,4].

A considerable number of studies has then investigated the different key factors responsible for stimulating longer time spent under high oxygen fractions during short-duration effort HIIT [15-60seconds] (e.g., speed intensity and the

relation between work and recovery) [5–10]. Intensities between 100 and 120% of the velocity associated with VO_{2max} (vVO_{2max}) or of the peak track tests or treadmill tests velocity have already been described as relevant criteria to the characterization and analysis of this exercise stimulus [6,7,9]. On the other hand, the relationship between work and recovery (W/R relation) remains unclear, due mainly to the differing results found so far [5,8,10], therefore suggesting that new analyses should be made before projecting such outcomes onto HIIT sessions prescription.

Additionally, any adjustments to work and/or recovery durations may directly impact training session volume (total exercise time), which is a fundamental aspect for the stimulation of longer time spent under high fractions of oxygen uptake during short-duration HIIT [3]. In theory, a better adjustment to the W/R relation would optimize a longer time at or above 90% of VO_{2max} , as it allows greater exercise volumes per session.

Thus, the aim of the present study was to verify the training session volume and the time spent at or above 90% of VO_{2max} under different work and recovery settings during the HIIT.

2. Materials and methods

2.1. Subjects

Eleven amateur male runners were recruited to the study in an intentional non-probabilistic manner (age: 25.45 ± 4.13 years; stature: 174 ± 0.09 cm; body mass: 64.91 ± 8.22 kg, body mass index 21.47 ± 1.52 kg.m⁻² and personal best in urban 5 km: 18 min: 30 sec $\pm$ 01 min: 25 sec). All men were healthy, with over two years of running experience, did not present any recent musculoskeletal injury or cardiorespiratory problems, and did not use any performance-altering substance. The study was approved by the local ethics committee and performed according to the Declaration of Helsinki.

2.2. Study design

All individuals executed one incremental test session and three other experimental sessions along the two weeks of testing. The volunteers were told not to practice any vigorous physical activity during the 24 hours before tests and not to have energetic or caffeinated drinks in the 12 hours prior to them. The order of the experimental sessions was randomized and counterbalanced, with intervals of 72–96 hours between them. All tests occurred at the same time of day in a room with controlled atmospheric conditions (temperature range: 22–24 °C and humidity: 50–60%).

2.3. Maximal incremental exercise test and determination of peak velocity

The maximal incremental exercise test was performed in a motor-driven ergometric treadmill (Movement, E-740–Brazil) with inclination fixed at 1%. After a three-minute warming at 6 km.h⁻¹, athletes began incremental protocol with a starting treadmill speed of 9 km.h⁻¹. A 30-second passive interval occurred every two minutes, right followed by a speed increase of 1 km.h⁻¹ [11], which allows the volunteer to reach a V_{peak} greater than the continuous [12] form without altering the VO_{2max} values [11], in addition to approaching the experimental condition, whose protocols were performed in a discontinuous way. The heart rate was monitored continuously throughout exercise using a COSMED heart rate monitor (QUARK PFT ERGO model, Italy). The rating of perceived exertion (RPE) was measured between each break for speed increment, using a 6-20 pt. scale [13]. A portable lactimeter (ROCHE, Accutrend® Pluss, Basel, Switzerland) was used to gauge the blood lactate levels after four minutes the incremental test [11].

The V_{peak} was assumed based on the speed at the last completed workload. For the incomplete workloads, velocity was readjusted according to Kuipers et al. (1985) [14] recommendations, in which: V_{peak}=velocity at the last complete workload + [(time (seconds), sustained during the not completed workload/120) * velocity increment (1 km.h⁻¹)].

2.4. Determination of maximal oxygen uptake (VO_{2max})

During the incremental test was used an ergospirometer to measure the breath-by-breath oxygen uptake (VO₂) from the exhaled gas (Quark PFTergo, Cosmed, Srl, Rome, Italy). The analysis system was calibrated prior to each evaluation, as recommended by the manufacturer.

The VO₂ values were reduced in intervals of 30 seconds, with the highest recorded value being considered as maximal. Some measures were taken in order to consider oxygen uptake to be maximal, such as: respiratory exchange ratio (RER) equals to or above 1.1; heart rate $\geq 90\%$ of age-predicted rate [$206 - (0.7 \times \text{age})$]; and a VO₂ plateau (< 150 ml.min⁻¹) after incrementing speed [15,16].

2.5. High-intensity intermittent protocols

Volunteers had a five-minute warming at 65% of V_{peak} during all protocols, followed by two five-second series at 100% of V_{peak} with 30 seconds of break between them. Experimental protocol was started three minutes after warming had finished.

The experimental sessions were performed with 40 seconds of work and 20 or 40 seconds of passive recovery (40s/20s and 40s/40s, respectively), and with 30 seconds of work and 30 seconds of passive recovery (30s/30s), with treadmill inclination at 1%. Verbal encouragement was given during the test. The ended the test when it could not be completed under the stipulated time for work complement (30 or 40 seconds).

The session volume was divided in: total exercise time (TT), which combines work and recovery time; number of complete series performed (NS); and time of exercise (TE), which excludes recovery time.

2.6. Time spent at or above 90% of VO_{2max}

The same oxygen uptake system as of the incremental test was used during the experimental sessions (Quark PFTergo, Cosmed, Srl, Rome, Italy). However, breath-by-breath values were reduced to the average of five seconds. The values found in the experimental sessions were determined as the time in which the VO₂ and values stayed at or above 90% of the maximal values from incremental test, as cited by Midgley et al. (2007) [11].

2.7. Analyses of blood lactate concentrations

The portable lactimeter (ROCHE, Accutrend® Pluss, Basel, Switzerland) was used to gauge the blood lactate levels. After discarded the first blood drop after puncture to avoid contamination from alcohol and/or sweat were collected blood drops to fill the reactive strip. This procedure was executed and three times in each experimental protocol (immediately after; five minutes after and ten minutes after).

Table 1 Description of the NS, TT and TE.

Variables	Protocols		
	40/40	30/30	40/20
NS (n)	12.91 ± 0.87 (10.97–14.84) ^a	24.18 ± 1.48 (20.89–27.47) ^b	6.72 ± 0.51 (5.6–7.85)
TT (s)	1043 ± 69.39 (888.2–1197) ^a	1463 ± 86.14 (1271–1655) ^b	415.55 ± 30.08 (348.5–482.6)
TE (s)	526.45 ± 34.72 (449.1–603.8) ^a	737.36 ± 41.9 (644–830.7) ^b	281 ± 20.01 (236.4–325.6)

Mean ± standard error mean (CI95%); number of complete series performed (NS); total exercise time (TT); time of exercise (TE).

^a Significant difference of 40:20 protocol ($P < 0.05$).

^b Significant difference of 40:40 and 40:20 protocols ($P < 0.05$).

2.8. Statistical analyses

The data is presented as mean ± standard error mean (SEM) and confidence interval of 95% (CI95%). Normality was checked by using the Shapiro-Wilk test. In addition, the Mauchly test was used to check for sphericity, and the Greenhouse-Geisser adjustment was applied whenever this assumption was not valid. For the comparisons between the different cardiorespiratory variables and the volume of exercises session we used a variance test (ANOVA) for repeated measures. As for the lactate concentration, we performed a two-way test (protocols × times) with a Bonferroni post-hoc test whenever we found interactions between the variables. Significance was accepted at $\alpha < 0.05$ for all statistical analyses. As a complement, Cohen's effect size (d) was evaluated as trivial (< 0.19), small (0.2–0.59), moderate (0.6–1.19), large (1.2–1.99), very large (2.0–3.99) and nearly perfect (> 4.0).

3. Results

The V_{peak} , 110% of V_{peak} , HR_{max} , VO_{2max} , maximal respiratory exchange ratio, lactate concentration and RPE for volunteers during incremental test were $18.15 \pm 0.25 \text{ km} \cdot \text{h}^{-1}$, $19.96 \pm 0.28 \text{ km} \cdot \text{h}^{-1}$, $196 \pm 2.62 \text{ bpm}$, $55.88 \pm 1.43 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$, 1.38 ± 0.06 , $11.65 \pm 0.61 \text{ mmol} \cdot \text{L}^{-1}$ and $19.64 \pm 0.20 \text{ pts.}$, respectively.

3.1. Number of complete series performed, total exercise time, and time of exercise

The largest number of complete series performed by volunteers was reached in the 30s/30s protocol, which was statistically different from protocols 40s/40s ($P = 0.0004$; $d = 2.87$) and 40s/20s ($P < 0.0001$; $d = 4.76$). Protocols 40s/40s and 40s/20s also presented significant difference between each other ($P = 0.0002$; $d = 2.62$). Were found the same pattern for total exercise time, which was longer for the 30s/30s protocol compared to the 40s/40s ($P = 0.01$; $d = 1.61$) and to the 40s/20s one ($P < 0.0001$; $d = 4.89$), and with protocol 40s/40s presenting more executed series than the 40s/20s one ($P < 0.0001$; $d = 3.53$). Likewise, protocol 30s/30s presented longer time of exercise than both 40s/40s ($P = 0.01$; $d = 1.65$) and 40s/20s ($P < 0.0001$; $d = 4.19$) protocols, with protocol 40s/40s being longer than protocol 40s/20s ($P = 0.0002$; $d = 2.61$). Table 1 presents the mean,

the mean standard error and the CI 95% values obtained from these variables.

3.2. Blood lactate concentrations

The blood lactate levels in millimoles per liter immediately after, five minutes after, and ten minutes after protocols are shown in Table 2. There was significant difference between protocols 40s/20s and 30s/30s ten minutes after ($P = 0.01$; $d = 0.98$).

3.3. Time spent at or above 90% of VO_{2max}

There was no statistical difference between protocols in $t90VO_{2max}$ ($F = 0.36$; $P = 0.69$) (Fig. 1). On the other hand, the maintenance values at or above 90% of VO_{2max} in relation to total exercise time ($t90VO_{2max}$ (%TT)) demonstrated statistical differences in the HIIT ($F = 16.21$; $P < 0.01$). In $t90VO_{2max}$ (%TT), the 40s/20s protocol differed from the others and had a bigger percentage of time ($48.33 \pm 5.8\%$) than protocols 40s/40s ($22.62 \pm 5.41\%$; $P = 0.01$; $d = 4.58$) and 30s/30s ($15.94 \pm 4.09\%$; $P = 0.003$; $d = 6.45$).

4. Discussion

Our results did not present any significant difference between protocols with different W/R at raw $t90VO_{2max}$. Quantitative variables of exercise volume were, however, significantly different when distinctive W/R settings were prescribed. So, by showing that the work and recovery ratios may unusually effect the cardiorespiratory and volume responses, we expand the understanding of the relationship between the prescription variables (W/R) and the performance parameters (cardiorespiratory ratios).

4.1. Number of series executed, total exercise time, time of exercise and lactate levels.

The results here presented point to a difference between protocols, with the greatest volume of exercise found for the 30s/30s protocol. Interestingly, adding 10 seconds to the work and recovery (40s/40s) led to the reduction of 28% in time of exercise. This result corroborates with Assadi & Lepers (2012) [5], who similarly found alterations in time of exercise after increasing work and recovery mutually.

Table 2 Description of the lactate concentration values.

Moments	Protocols		
	40/40	30/30	40/20
Immediately after	11.4 ± 0.49 (10.3–12.46) ^a	11.4 ± 0.75 (9.72–13.07) ^a	11.9 ± 0.84 (9.98–13.75)
5 minutes after	9.6 ± 0.41 (8.65–10.47)	9.2 ± 0.59 (7.92–10.57) ^b	11.2 ± 1.03 (8.87–13.45)
10 minutes after	8.1 ± 0.55 (6.87–9.3)	7 ± 0.49 (5.87–8.03)	9.9 ± 1.17 (7.26–12.48) ^c

Mean ± standard error mean (CI95%).

^a Significant difference between the moments after ($P < 0.05$).

^b Significant difference between the moment 10 minutes after ($P < 0.05$).

^c Significant difference of protocol 30:30 ($P < 0.05$).

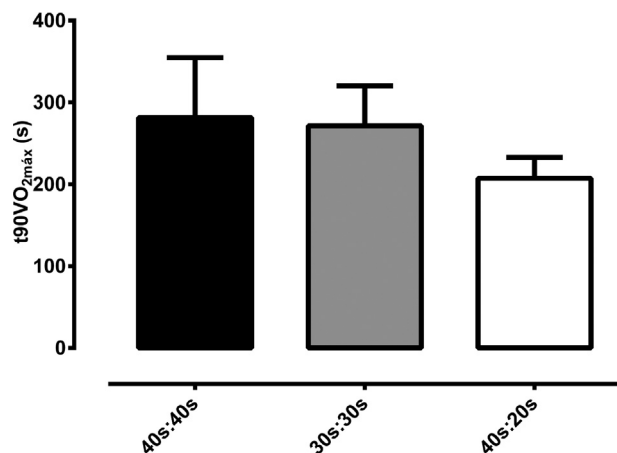


Figure 1 The time spent at or above 90% of absolute VO_{2max}.

Considering this, it may be inferred that the increase in work time (from 30 to 40seconds) resulted in a greater energetic necessity of keeping the exercise, and even an increase in the recovery time (also from 30 to 40seconds) could not raise the exercise final duration. The metabolic acidosis caused by the time spent at high intensity associated with a longer work time [17] may have harmed the phosphocreatine (PCr) resynthesis, thus delaying the availability of fast energy for new concentrations [18,19] and impairing the exercise continuity.

Besides, the number of series performed and the time of exercise in the session were halved after work time was kept and recovery time was reduced (40s/40s to 40s/20s). As for the total exercise time this value increased almost threefold. During the intermittent training, passive recovery periods are characterized by the fast reduction of oxygen uptake. In such periods oxygen is readily reset in tissues, with most of the depleted adenosine triphosphate (ATP) and the PCr being restored in the muscles. It is believed that a little more than 70% of phosphagens is restored within 30seconds [18,20]. Thus, the halving of recovery time led to an improper restoration of the ATP and PCr stocks with a probable accumulation of metabolites, which are responsible for distinctive volumes between these protocols.

It is important to highlight that HIIT requires the participation of the three energy attainment systems (ATP-PCr; glycolytic and oxidative) [18,21,22]. In spite of this contribution being mostly anaerobic (ATP-PCr and glycolytic), the aerobic components (oxidative) also contributed to

work maintenance by providing energy as ATP during 30 to 90seconds work [22].

High lactate concentrations emphasize the great energy contribution from the anaerobic metabolism for both protocols [10]. In our study, high lactate levels persisted for 10minutes during the 40s/20s protocol. This point was the only one able to show a difference to another protocol (30s/30s).

One possible explanation is that a high lactate production persisted because of muscle fibers that possibly provided poor quantities of oxygen, thus sustaining the signals of the glycolytic pathway activity with PCr degradation and lactate production [23]. Another hypothesis is the increase in the lactate levels accumulation speed with a consequent reduction in the capacity of tamponade by the saturation of intracellular mechanisms responsible for removing lactate [24], such as monocarboxylate transporters and lactate dehydrogenase [25]. However, studies on HIIT and the lactate levels right after training are necessary for this hypothesis to be confirmed.

4.2. Effect of time spent at or above 90% of maximal oxygen uptake on different ratios between work and recovery.

Our results show that oxygen fractions at or above 90% of maximal value behave in the same way when different relations of W/R until exhaustion are prescribed for running trainings.

It is important to point that the maximal and supramaximal intensities (100 to 120%) have already been described as excellent stimuli to boost the t90VO_{2max} during running [6,7,9]. Yet, studies performed on treadmills and tracks have presented various responses to the variable when W/R was manipulated, with studies both showing difference between work and recovery ratios [8,10] or not [5]. We must also emphasize that the studies that found differences between protocols were developed under a fixed distance or according to the exhaustion time.

Thus, this study reinforces that HIIT (until exhaustion is reached) with fixed intensity, does not modify fractions superior to 90% of the oxygen uptake when different ratios between effort and recovery are prescribed, similarly to another study available in literature [5].

In the present study, during the 40s/20s protocol, recovery time allowed lower reduction of oxygen. In sequence (next work), the subjects reached again fractions equal or

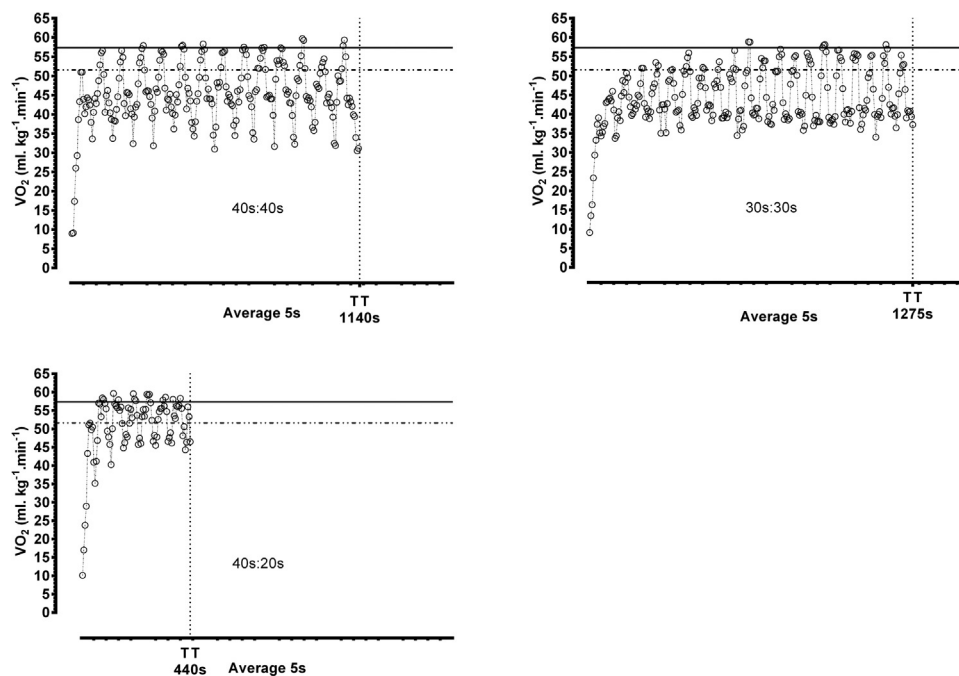


Figure 2 Example of VO_2 behavior over time.

superior to 90% of the oxygen consumption. However, with depletion of oxygen reserves, and higher usage of the glycolytic system, the accumulation of metabolite (inorganic phosphate, hydrogen ions and potassium) occurs, reducing exercise tolerance and preventing higher oxygen fractions from staying a longer period.

By increasing recovery time, though, as in the 40s/40s case, accented reduction of oxygen and heart rate occurs, attenuating the values of these variables during the subsequent efforts. Besides, as a compensatory effect, the oxygen consumption kinetics is accelerated when the recovery interval is longer, allowing the oxygen fractions to return to higher values during the exercise sessions, without modifying the energetic contribution system [26,27].

Despite the absence of studies in severe domains investigating different ratios of effort and recovery, other studies in heavy domains consider these observations mentioned above (the oxygen fluctuation and the oxygen consumption kinetics) [26,28]. In fact, the 30s/30s protocol may generate higher oxygen fluctuations during the session, with enough time for a good recovery of the necessary energy (PCr resynthesis and restoration of the oxygen stacks in the myoglobin). Consequently, the metabolic perturbation is lower, conducting the exercise during a longer period when compared to others, but with oxygen fractions impaired by the excessive number of fluctuations.

It should also be highlighted that, when considering the $t_{90\text{VO}_{2\text{max}}}$ in percentage of total exercise time, protocol 40s/20s presented higher levels were than the two other ones. Thus, it is viable to say that the protocol choice may optimize training when prescription intention is to emphasize the different aspects of oxygen uptake, as $t_{90\text{VO}_{2\text{max}}}$ percentages in relation to total exercise and exhaustion time were higher for the protocol with shorter duration (40s/20s). This fact mainly owes to the oxygen fluctuations (Fig. 2) and physiological conditions discussed above.

Thus, it is suggested that both propositions of ratio between work and recovery used in our study may be used during the preparatory period of the athlete periodization due to the importance of this stage to create bases for physical conditioning as a $\text{VO}_{2\text{max}}$ improvement, they may also be used during the shock microcycle, aiming to increase the stress of the training [29].

5. Conclusion

In concluded that the studied protocols (30s/30s, 40s/40s and 40s/20s) at 110% of V_{peak} until exhaustion presented approximated $t_{90\text{VO}_{2\text{max}}}$ levels, with no difference between them. They also presented distinctive volume variables, with protocol 30s/30s presenting a longer time of exercise than the two others. From this, we may use the present results for the prescription of training sessions, with the 30s/30s protocol being the most appropriate when there is the goal to increase the volume session. On the other hand, protocol 40s/20s may optimize training time when the purpose is to stimulate high oxygen fractions, as its results are like those of other protocols in $t_{90\text{VO}_{2\text{max}}}$ with time reduced up to four times.

Disclosure of interest

The authors declare that they have no competing interest.

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